Last time. Proving $\Gamma(G/B, i_{w*}\mathcal{O}_{X_w}) \simeq \nabla_w$ ← $\mathfrak{g}$-actions

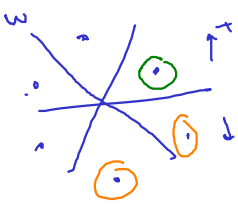
Remaining: characters agree — equivariant for use the left H-action, $\Rightarrow$ H-action on global sections — H-action

$$X_w = BwB/B \subseteq G/B \xhookrightarrow{\text{cl}} Y_w = wU^-B/B \xhookrightarrow{\text{op}} X = G/B.$$

- Bruhat cells $X_w = UwB/B$ b/c w normalizes T $\quad wT = Tw$
- every Bruhat orbit has unique T-fixed point: $wB/B = x_w \in X_w$
- "generic" X_w using action of $U_w \subseteq U$ on $x_w = wB/B$ action transitive $\Longrightarrow$ $\boxed{X_w \simeq U_w \simeq \mathbb{A}_w}$

α root $\rightsquigarrow$ $U_\alpha \simeq \mathbb{G}_a \subseteq G$ $\quad$ Notation: $S \subseteq$ roots, $U_S = \prod_{\alpha \in S} U_\alpha$ (product)

$U = U_{\Delta^+}$ $\quad$ Question: when does U_α stabilize wB/B?



Check: $U_\alpha w = w(w^{-1}U_\alpha w) = w\,U_{w^{-1}(\alpha)}$

$\Longrightarrow$ $U_\alpha wB/B = w\,U_{w^{-1}(\alpha)}B/B = wB/B$ iff $w^{-1}\alpha \in \Delta^+$, $\alpha \in w(\Delta^+)$ (stabilizer)

$U_w = \prod U_\alpha$ where $\boxed{\begin{array}{l}\alpha \in \Delta^+ \\ \alpha \in w(\Delta^-)\end{array}}$ i.e. $\alpha \notin w(\Delta^+)$

$\cap$ U

Sanity check: $X_w \simeq U_w$ 1 dimensional, $\ell(w) = 1$

- Study $Y_w = wU^-B/B$.
  - contain X_w i.e. $U_wB/B \subseteq wU^-B/B$
  - automatically affine b/c U^- is
  - maximal dimension: "big cell" $\simeq U^-B/B$, U^- stabilize eB/B. $\dim = \dim(U^-) = \dim(G/B)$

Claim: $U^-_w \subseteq U^-$ subgroup acts transitively in Y_w

- $U^-_w \cdot X_w = Y_w$

$U^-_w \times X_w \xrightarrow{\simeq} Y_w$ $\quad$ realizes $X_w \xhookrightarrow[\text{closed}]{} Y_w$

$\{e\} \times X_w \hookrightarrow$

Restrict attention to wB/B $\qquad U_\alpha wB/B = wU_{w^{-1}(\alpha)}B/B$. $\qquad w^{-1}(\alpha)\in\Delta^+$, $\alpha\in w(\Delta^+)$

U_α fixes wB/B iff: $\alpha\in\Delta^-\cap w(\Delta^+)$

complement: $U_{\Delta^-\cap w(\Delta^-)} = U_w^-$ $\qquad$ • $U_w\cap U_w^-$ disjoint

$U_{\Delta^+\cap w(\Delta^-)} = U_w$ $\qquad$ • union is $U_{w(\Delta^-)}$

$\Rightarrow U_{w(\Delta^-)}\cdot(wB/B)$ maximal dimensional b/c $= wU^-B$.

$\dim(U_{w(\Delta^-)}) = \dim(G/B)$

Summary: $X_w = U_w wB/B \xhookrightarrow[\text{cl.}]{i} Y_w = U_w^- U_w\, wB/B \xhookrightarrow[j]{\varphi} G/B$.

Want: $\underset{\text{Opsh}}{\partial}\, \underset{\text{D-mod.}}{i_*}\mathcal{O}_{X_w}$ $\qquad$ filter $i_*\mathcal{O}_{X_w}$ such that graded is:

$\mathrm{Sym}_{\mathcal{O}_{X_w}}(\mathcal{N}_{X_w/Y_w})\otimes\det(\mathcal{N}_{X_w/Y_w})$ "pushed"

$\omega_{X_w}\otimes\omega_{Y_w}^{-1}$

left $\xrightarrow[\otimes\omega_{X_w}]{}$ right $\xrightarrow[\text{push}]{}$ right $\xrightarrow[\otimes\omega_{Y_w}^{-1}]{}$ left

$0\to\mathcal{N}^\vee_{X_w/Y_w}\to\Omega^1_{Y_w}\to\Omega^1_{X_w}\to 0$

$\det(\mathcal{N}^\vee) = \omega_{Y_w}\otimes\omega_{X_w}^{-1}$

$\det(\mathcal{N}) = \omega_{Y_w}^{-1}\otimes\omega_{X_w}$

$\mathcal{O}_w = \mathcal{O}(X_w) = \mathcal{O}(U_w) = \mathrm{Sym}(U_w^*)$

$\mathcal{N}_{X_w/Y_w}\simeq U_w^-$

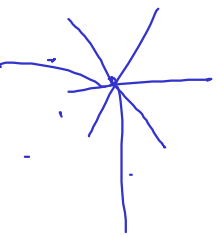
$\rightsquigarrow \mathrm{Sym}(U_w^*\oplus U_w^-)$

$\uparrow$ Δ^-

U: U_w^-: $\Delta^-\cap w(\Delta^-)$

U_w: $\Delta^+\cap w(\Delta^-)$

U_w^*: $\Delta^-\cap w(\Delta^+)$



$= \mathrm{Sym}(U^-)$

Wanted: character $=$ character $(\nabla_{w(-2\rho)}) = \mathrm{ch}(\Delta_{w\cdot(-2\rho)})$

$w\cdot(-2\rho) = w(-\rho)-\rho = \frac{1}{2}(w(\Sigma\,\text{neg})-\Sigma\,\text{neg})$

Show: $\det(U_w^-) =$ sum of roots in $\Delta^-\cap w(\Delta^-)$

$=$ sum neg roots

Riemann-Hilbert: $k=\mathbb{C}$

Before: define dR functor: $dR_X(M) = \Omega^\bullet_X \otimes_{\mathcal{O}_X} M$

$$\omega\otimes m \mapsto d\omega\otimes m + \sum \omega\, dx_i \otimes \frac{\partial}{\partial x_i} m$$

enough: $\Gamma(X, dR_X(M))$.

· observe that differential is not $\mathcal{O}_X$-linear. e.g. $M=\mathcal{O}_X$, dR diff.

$\Rightarrow dR_X(M)$ lives in weird category: expect: Γ not exact on affine.

Example: $X=\mathbb{A}^1$ $\quad M = D_X/D_X(\partial-1) \underset{\partial}{\cong} \mathcal{O}_X e^x$ "generated by e^x"

$\frac{d}{dx}f - f = 0$

$f = e^x$

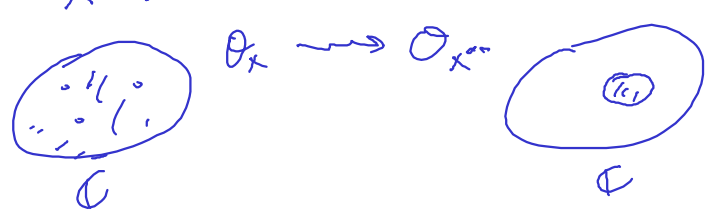
$$dR_X(M) = \mathcal{O}_X \xrightarrow[f \mapsto (f+f')dx]{} \mathcal{O}_X\, dx \qquad fe^x \mapsto (fe^x)'\,dx = f'e^x + fe^x$$

Check: · acyclic on global sections

· on $\mathcal{U}$, it has cokernel: $\frac{1}{x}e^x dx$ $\quad \frac{1}{x}dx$

Filtration: analytification: X scheme $\rightsquigarrow X(\mathbb{C})$ w/ analytic topology.

$\mathcal{O}_X \rightsquigarrow \mathcal{O}_{X^{an}}$



Ex: analytification.

$$\mathcal{O}_X \xrightarrow[f \mapsto f+f']{} \mathcal{O}_X$$

Check: using power series

· $f+f'=g$ has solution locally.

· has kernel: $f(x) = e^{-x}$

$dR(M) = \mathbb{C}_{\mathbb{C}^{an}}$

$dR^{an}_X(M) := dR_X(M)^{an} \in DSh(X^{an}; \mathbb{C})$

Question: is this constructible, etc....